EXP 2

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YOLO11 (You Only Look Once version 11) is a real-time object detection model that can detect multiple objects in images and videos in a single forward pass of a neural network. Unlike traditional methods that scan an image many times, YOLO looks at the entire image only once, making it extremely fast and efficient.

YOLO11 is widely used in applications where speed and accuracy are both important, such as surveillance systems, traffic monitoring, autonomous vehicles, sports analytics, medical imaging, and smart retail systems. It can detect objects like people, vehicles, animals, bags, tools, and many more in real time.

YOLO11 uses deep convolutional neural networks and optimized architectures to provide better accuracy, faster inference speed, and lower computational cost compared to older YOLO versions.

Its models are such that

Model	Description	Speed	Accuracy	Best Use
YOLO11n (Nano)	Smallest and fastest model	Very Fast	Medium	Low-end devices, realtime apps
YOLO11s (Small)	Lightweight balanced model	Fast	Good	Laptops and edge devices
YOLO11m (Medium)	Balanced accuracy and speed	Medium	High	GPUs and research
YOLO11l (Large)	High accuracy model	Slower	Very High	Industrial systems
YOLO11x (Extra Large)	Highest accuracy model	Slowest	Maximum	High-end GPUs

USES:

YOLO11 is used in many real-world applications:

1. **Traffic Monitoring**
   Detects vehicles, number plates, congestion, and accidents.
2. **Security & Surveillance**
   Detects people, suspicious activity, intrusion, and crowd counting.

3. **Autonomous Vehicles & Drones**

Detects pedestrians, vehicles, obstacles, and road signs.

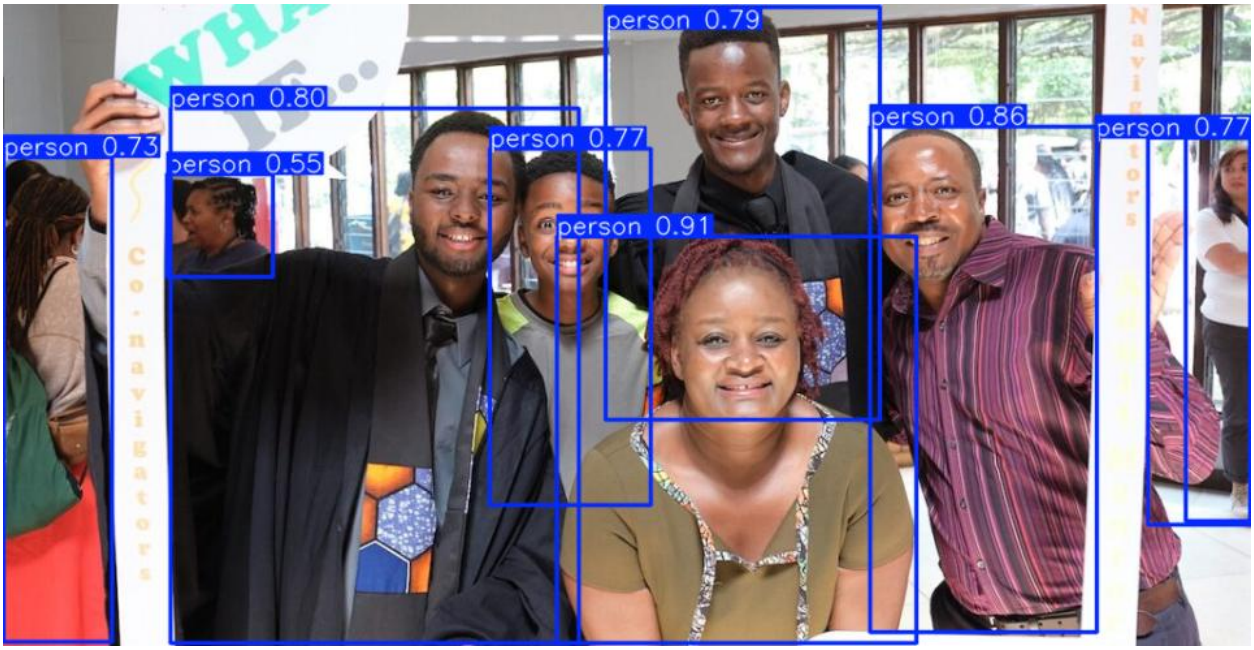
4. **Retail Analytics**
   Customer counting, product tracking, theft detection.
5. **Healthcare**
   Detects medical objects in scans and lab automation.
6. **Sports Analytics**
   Player tracking, ball detection, motion analysis.\

CODING EXERCISE:

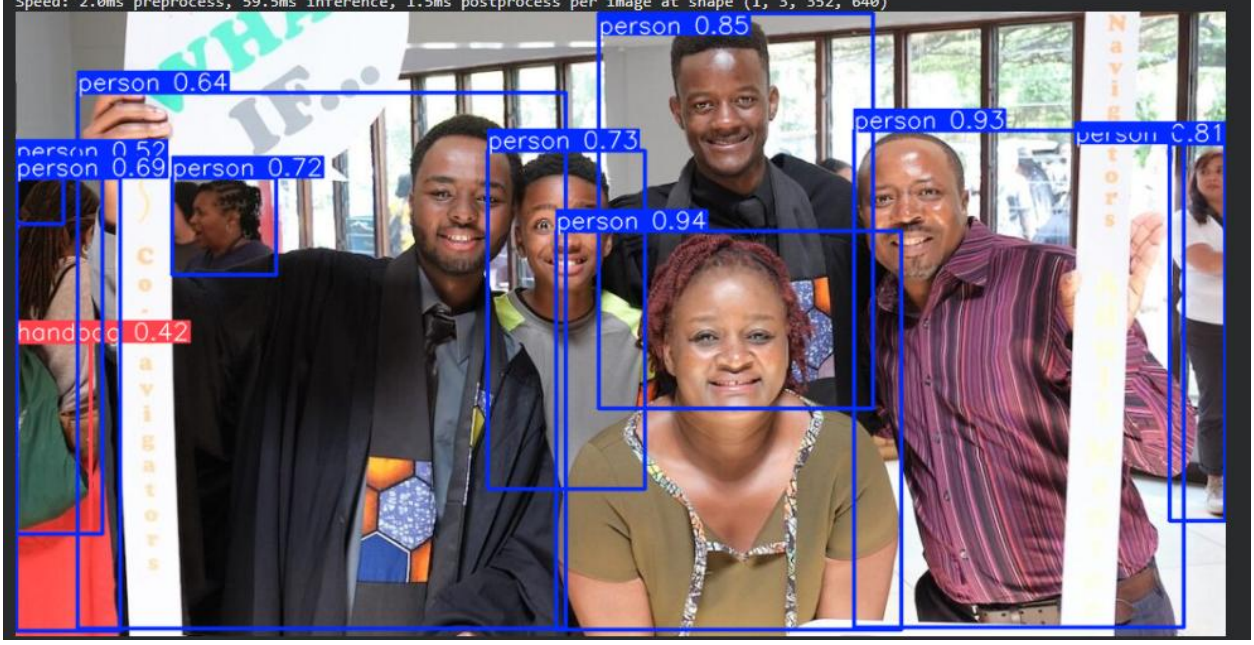
1. For image:

```
image_path = "/content/kenya-265-1024x528.jpg"

person_confidence = {}

for name, model in models.items():
    print(f"\nRunning {name} on image")

    results = model(image_path, conf=0.4)
    boxes = results[0].boxes

    max_person_conf = 0.0

    if boxes is not None:
        for box in boxes:
            cls_id = int(box.cls[0])
            conf = float(box.conf[0])
            class_name = results[0].names[cls_id]

            if class_name == "person":
                max_person_conf = max(max_person_conf, conf)

    person_confidence[name] = max_person_conf
    print(f"Highest person confidence in {name}: {max_person_conf:.3f}")
```

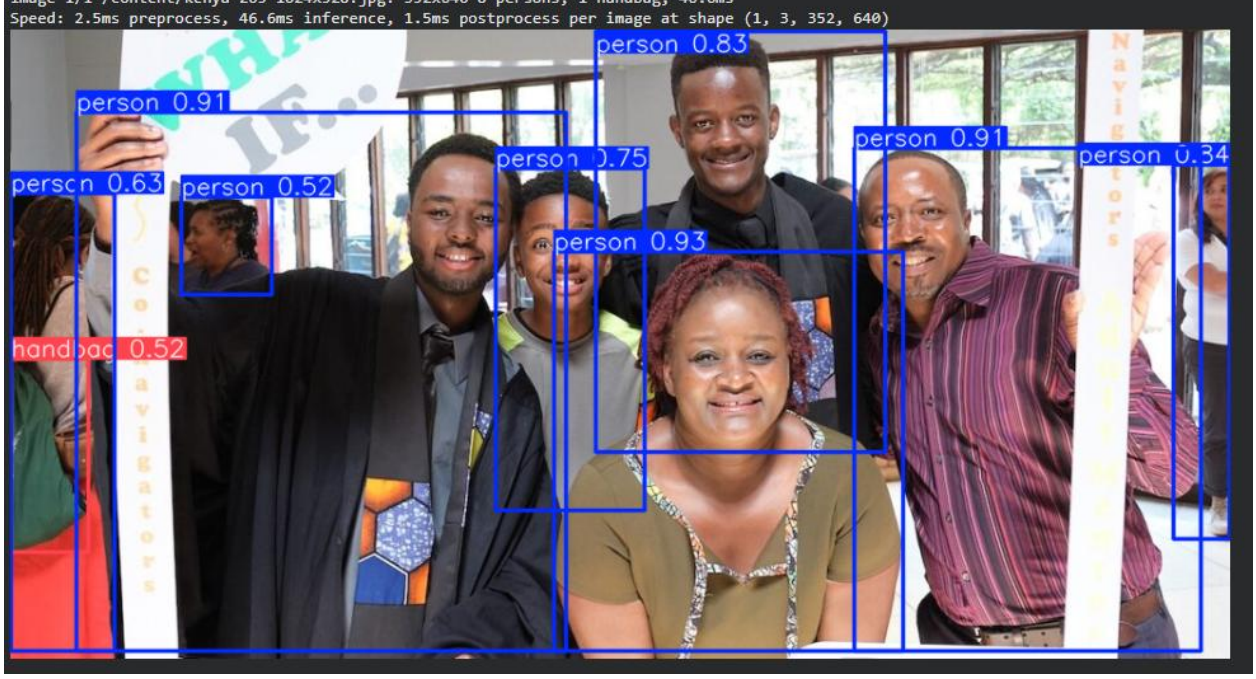
```
Running YOLOv11n on image

image 1/1 /content/kenya-265-1024x528.jpg: 352x640 9 persons, 10.0ms
Speed: 2.5ms preprocess, 10.0ms inference, 1.5ms postprocess per image at shape (1, 3, 352, 640)
Highest person confidence in YOLOv11n: 0.911

Running YOLOv11s on image

image 1/1 /content/kenya-265-1024x528.jpg: 352x640 10 persons, 1 handbag, 10.9ms
Speed: 1.6ms preprocess, 10.9ms inference, 2.0ms postprocess per image at shape (1, 3, 352, 640)
Highest person confidence in YOLOv11s: 0.940

Running YOLOv11m on image

image 1/1 /content/kenya-265-1024x528.jpg: 352x640 8 persons, 1 handbag, 23.3ms
Speed: 1.6ms preprocess, 23.3ms inference, 1.1ms postprocess per image at shape (1, 3, 352, 640)
Highest person confidence in YOLOv11m: 0.931

Running YOLOv11l on image

image 1/1 /content/kenya-265-1024x528.jpg: 352x640 8 persons, 1 handbag, 29.9ms
Speed: 1.5ms preprocess, 29.9ms inference, 1.1ms postprocess per image at shape (1, 3, 352, 640)
Highest person confidence in YOLOv11l: 0.950

Running YOLOv11x on image

image 1/1 /content/kenya-265-1024x528.jpg: 352x640 8 persons, 1 backpack, 62.2ms
Speed: 1.6ms preprocess, 62.2ms inference, 1.2ms postprocess per image at shape (1, 3, 352, 640)
Highest person confidence in YOLOv11x: 0.953
```



```
Running YOLOv11s on image

image 1/1 /content/kenya-265-1024x528.jpg: 352x640 10 persons, 1 handbag, 59.5ms
Speed: 2.0ms preprocess, 59.5ms inference, 1.5ms postprocess per image at shape (1, 3, 352, 640)
```




```
image 1/1 /content/kenya-265-1024x528.jpg: 352x640 8 persons, 1 handbag, 46.6ms
Speed: 2.5ms preprocess, 46.6ms inference, 1.5ms postprocess per image at shape (1, 3, 352, 640)
```



This code uses the ultralytics library to perform object detection on an image using three different YOLO models. For each model, it loads the model, runs inference on people.jpg, displays the results, and saves the annotated output to a separate .jpg file.

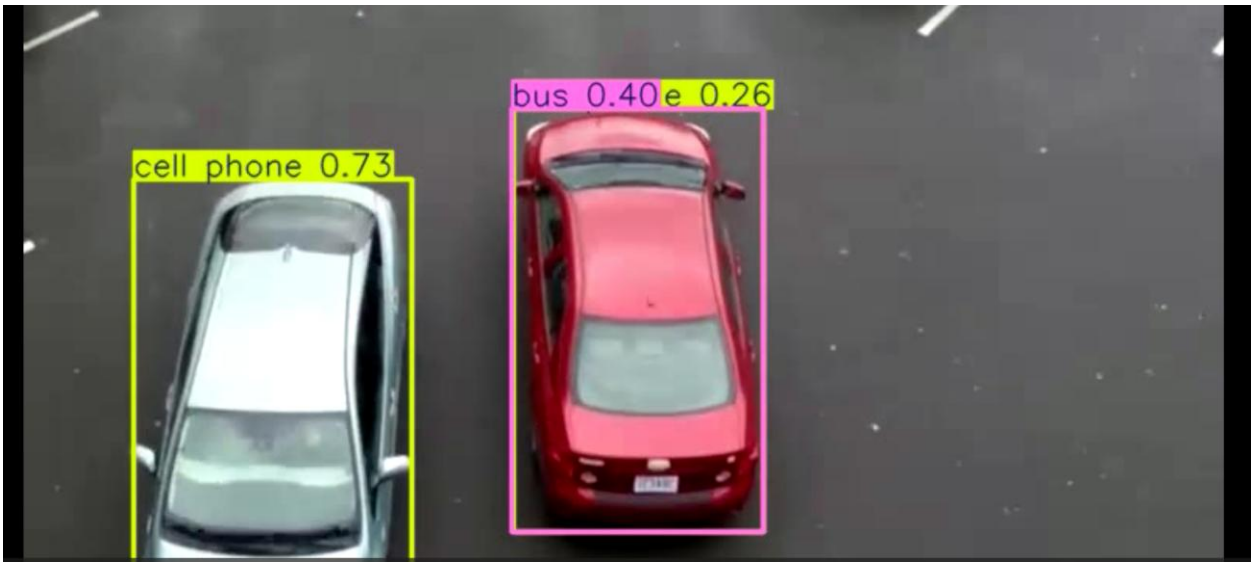
- **For highest detection (e.g., more persons, cars, traffic lights identified):** yolo11x.pt, yolo11s.pt, and yolo11l.ptdetected the most 'persons' (10 persons each).
- **For fastest inference speed:** yolo11n.pt was the fastest at 112.2ms

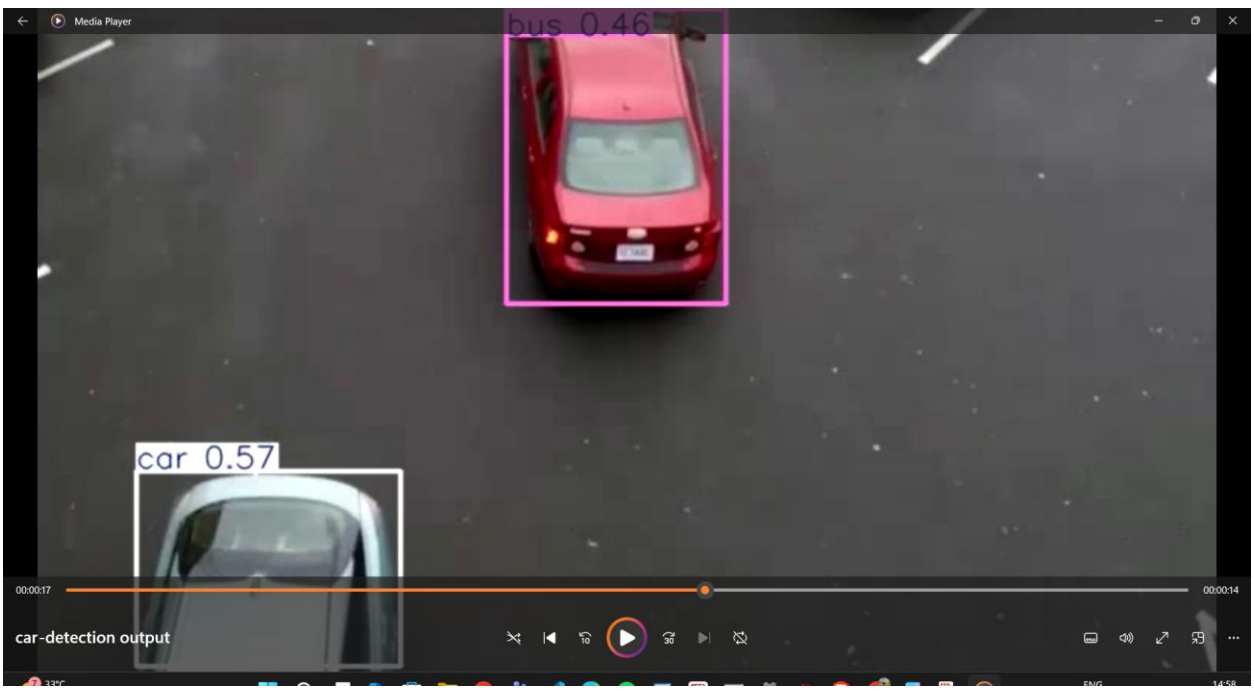
  One of the output is:

We get 5 such outputs in total

For Video:

```
video_path = "/content/bolt-detection.mp4"

for name, model in models.items():
    print(f"\nRunning {name} on video")
    model.predict(
        source=video_path,
        save=True,
        conf=0.4,
        project="yolo11_outputs",
        name=name
    )
```





This code iterates through a list of YOLO models (yolo11n to yolo11x). For each model, it loads the model and then processes a video file (video.mp4) frame by frame. It performs object detection on each frame, annotates the frame with the detection results, and then saves the annotated frames into a new output video file within the outputs directory.

- **Fastest inference speed:** yolo11n.pt is the fastest with an average inference speed of 118.2ms per image, followed by yolo11s.pt at 284.0ms per image.

- **Slower but potentially more accurate models:** yolo11m.pt and yolo11l.pt have significantly higher inference times (720.2ms and 820.2ms respectively). While the full output for yolo11x.pt is truncated, it is generally the largest and slowest model, implying it would offer the most comprehensive detection, similar to its performance on images.